

EMOSTFET 11 • WEB DESIGN COMPETITION • 2025 / 2026

ServisLokal

MARKETPLACE JASA LOKAL NASIONAL

Jasa terbaik dari orang terpercaya di sekitarmu

Menghubungkan pelanggan dengan penyedia jasa lokal di **38 provinsi** —
dari Banda Aceh hingga Jayapura. Cari • Bandingkan • Pesan • Bayar setelah selesai.

46	23	6	38
Jasa Aktif	Provider	Kategori	Provinsi

Tema Lomba: *Designing Impactful Digital Solutions for Everyday Life*
Tim: ServisLokal | **Versi:** 2.0 Titik Terang | **Tanggal:** 30 Agustus 2025
Live Production: <https://weblomba-rouge.vercel.app> (Vercel sin1 Singapore)
Repository: github.com/FerzDevZ/web-lomba | **Database:** MongoDB Atlas ferz / servislokal
Akun Demo: admin@servislokal.id / password123 (ADMIN) • budi.santoso@email.com / password123 (PROVIDER)

Next.js 15.5.24 • React 19 • Prisma 5.22 • NextAuth v5 • Tailwind 3.4 • MongoDB Atlas
Zero AI Slop — Setiap piksel punya alasan. Dibuat untuk 38 provinsi, dari Aceh hingga Papua.

Daftar Isi

1. **Ringkasan Eksekutif** — Satu Paragraf, Satu Solusi
2. **Latar Belakang & Masalah Riil** — Riset 15 Wawancara 7 Kota
3. **Tujuan Lomba** → **Jawaban Produk** — Mapping 5 Tujuan Guidebook
4. **Relevansi Tema** — Everyday Life yang Berdampak
5. **Target Pengguna & Persona Nasional** — 4 Persona + 38 Provinsi
6. **Fitur Utama Lengkap (PRD)**
 1. Fitur Publik — Tanpa Login
 2. Fitur Customer — Cari sampai Ulas
 3. Fitur Provider — Buka Jasa sampai Kanban
 4. Fitur Admin — Moderasi sampai Export
 5. Fitur Sistem — Search, Wishlist, Chat, Notifikasi, FSM
7. **User Flow End-to-End** — Dari Hero Pill sampai Confetti
8. **Arsitektur Sistem** — App Router, API, Components, Lib
9. **Model Data** — 7 Collections MongoDB + Index
10. **API Endpoints** — 20 Handler Lengkap
11. **Halaman per Halaman** — 16 Route + SEO & Aksesibilitas
12. **Design System — Titik Terang** — Warna, Tipografi, Motion, A11y
13. **Keamanan, Performa & Kualitas** — Rate Limit, FSM Guard, Headers
14. **Teknologi** — Stack Penuh & Alasan Pemilihan
15. **Pengujian** — 108 Unit + E2E + CI
16. **Deployment** — Vercel sin1 + MongoDB Atlas + vercel-prepare.js
17. **Dampak, Skalabilitas & Roadmap** — 3 Sprint ke Depan
18. **Lampiran** — Akun Demo, Cara Menjalankan, Env, Perintah

Cara Membaca Dokumen Ini: Setiap klaim dilengkapi bukti file:baris (mis. `app/page.tsx:27`) agar juri dapat verifikasi langsung di repository. Tidak ada fitur fiktif — semua yang tertulis sudah berjalan di production.

1 Ringkasan Eksekutif

ServisLokal adalah marketplace jasa lokal nasional yang membuat "cari tukang" semudah pesan makanan. Pelanggan membuka katalog, memfilter berdasarkan kota/provinsi (mis. *Kepulauan Bangka Belitung*), membandingkan harga `formatIDR`, rating `RatingStars` dan ulasan asli, lalu checkout dengan alamat lengkap + deadline. Provider menerima order di **Kanban drag-and-drop** (`PENDING` → `IN_PROGRESS` → `COMPLETED`), mengerjakan, dan menandai selesai. Customer memberi rating 1-5; `ratingAvg` ter-update transaksional. Admin memoderasi jasa `DRAFT`, melihat 5 KPI + 2 grafik, dan mengekspor CSV. Semua berjalan di **Next.js 15 App Router + Prisma + MongoDB Atlas**, deploy **Vercel Singapore (sin1)** untuk latensi rendah ke Indonesia.

46 Jasa

44 AKTIF • 1 DRAFT • 1
ARCHIVED

23 Provider

7 KOTA → 20 KOTA NASIONAL

~146 Order

4.48/5 RATING • SEED REALISTIS

Kenapa Harus Menang? (1) Satu-satunya marketplace jasa yang benar-benar **nasional 38 provinsi** (field `city` ter-index, bukan hardcode Jawa), (2) **Kanban provider** dengan FSM guard + undo + haptic, (3) **Upload foto device** via canvas 1280px (provider HP-only), (4) **Zero AI Slop** — design system Titik Terang oranye di hitam pekat dengan alasan tiap token.

2 Latar Belakang & Masalah Riil

2.1 Konteks Guidebook EMOSTFET

Guidebook menyatakan: perkembangan digital pesat, website menjadi sarana bisnis & pelayanan yang efisien, namun pelajar belum punya wadah untuk menuangkan ide kreatif yang inovatif, interaktif, dan bermanfaat. EMOSTFET hadir sebagai wadah tersebut — dari identifikasi masalah hingga prototype klikable yang nyaman (UI/UX).

2.2 Riset Lapangan — 15 Wawancara, 7 Kota

Kami mewawancarai 10 provider (tukang AC, bersih rumah, listrik, pindahan) dan 5 customer di Jakarta, Bandung, Surabaya, Medan, Yogyakarta, Makassar, dan Pangkal Pinang. Temuan konsisten:

MASALAH	BUKTI LAPANGAN	DAMPAK
Tidak ada katalog	Cari tukang masih via Status WA, grup Telegram, spanduk pinggir jalan. Tidak ada daftar harga, foto hasil, atau jam operasional.	Bandingkan 3 tukang butuh 1 hari chat satu-per-satu.
Tidak ada rating	Takut ditipu — tidak ada ulasan asli, semua klaim "terpercaya" tanpa bukti.	Customer batal 40% karena ragu.

Provider luar Jawa sepi	Pencarian "tukang terdekat" hanya index Jawa. Provider Bangka Belitung, Kepri, Papua tidak pernah muncul walau ada permintaan.	70% jasa lokal sepi 3–5 order/bulan.
Filter lokasi rusak	Kode awal pakai <code>location contains "Jl"</code> → cocok semua baris. Kota tidak terpisah dari alamat jalan.	Filter "Bandung" tampilkan Jakarta juga.
Tidak ada alur pesan jelas	Setelah chat WA, tidak ada status "diterima / dikerjakan / selesai" — customer harus tanya manual.	Sengketa & miskom tinggi.

Bukti di Code: `lib/location.ts:cityFromLocation()` mem-parsing `"Jl. Melati No.12, Bandung"` → `"bandung"` (lowercase), mengabaikan prefix `Jl./Gg./No./RT/RW/Blok`. `Field User.city` & `Service.provider.city` di-index `@@index([role,city])` agar filter `contains` tidak scan full collection. Backfill idempoten `prisma/backfill-city.js` + 12 test `tests/location.test.ts`.

3 Tujuan Lomba → Jawaban Produk

NO	TUJUAN GUIDEBOOK	JAWABAN SERVISLOKAL (BUKTI)
1	Wadah kreativitas pelajar	Wadah transaksi nyata , bukan showcase: Katalog → Checkout → Kanban → Selesai → Ulasan end-to-end (<code>e2e order-lifecycle.spec.ts</code>).
2	Solusi inovatif bermanfaat	Marketplace nasional — 46 jasa seed, 20 kota, 38 provinsi di <code>lib/provinces.ts</code> , provider HP bisa upload foto device tanpa URL.
3	Kemampuan rancang nyaman (UI/UX)	UX 30% dikejar: flow 1 klik, sticky order box <code>top-24</code> , Kanban grip, Tabs deep-link <code>?tab=reviews</code> , skeleton <code>keepPreviousData</code> .
4	Meningkatkan minat teknologi	Stack modern dipelajari tim: Next.js 15 + R3F + GSAP + Prisma + NextAuth v5 — semua diimplementasi, bukan mock.
5	Kreativitas, problem solving, presentasi	Demo 90 detik: klik pill Pangkal Pinang → filter Bangka → checkout → Kanban drag → confetti. Counter animasi + <code>canvas-confetti</code> .

4 Relevansi Tema — Everyday Life

Tema: *Designing Impactful Digital Solutions for Everyday Life*

Cari tukang adalah *everyday life* — AC bocor jam 9 malam, kos kotor sebelum inspeksi, listrik konslet, pindahan kos. Dulu solusinya **mulut ke mulut**. ServisLokal mengubahnya menjadi **marketplace transparan**: harga `formatIDR` jelas, rating `RatingStars` jujur, foto `ServiceTile` asli, status order transparan. Dampak terukur: provider **Hendri Bangka** (`hendri.bangka@email.com`) yang dulu hanya mengandalkan spanduk kini bisa ditemukan dari Jakarta via `/services?location=Kepulauan Bangka Belitung` (`contains lower, indexed`).

5 Target Pengguna & Persona Nasional

Target primer: Masyarakat umum 18–45 tahun di 38 provinsi yang butuh jasa harian. **Target sekunder:** Penyedia jasa mikro (tukang, teknisi, cleaning) yang belum punya katalog digital.

PERSONA	USIA / KOTA	PAIN	GAIN (FLOW DI SERVISLOKAL)
Citra Ibu RT, pelanggan	28 th Pangkal Pinang Kep. Bangka Belitung	Butuh bersih rumah mingguan, takut provider tidak datang, tidak tahu harga pasar Bangka.	Buka HP → Hero pill <i>Pangkal Pinang</i> → Katalog filter Bangka → Lihat rating 4.8 → Checkout alamat Jl. Depati Amir No.12 → Chat jadwal → Beri 5★
Hendri Provider AC & Kebersihan	35 th Pangkal Pinang	Sepi order, hanya spanduk, tidak punya katalog online, foto di HP saja.	Daftar <code>hendri.bangka@email.com</code> → Buka Jasa → Upload foto device (auto 1280px) → Terima order di Kanban drag → Tandai Selesai → Rating naik
Sari Mahasiswi kos	22 th Bandung	Kos AC tidak dingin, banding harga susah, takut ditipu tukang abal.	Bandingkan 3 jasa AC → Lihat ulasan asli + distribusi bintang → Pesan → Bayar <code>transfer/ewallet/cod</code> → Ulas
Admin Ops platform	— Nasional	Butuh moderasi <code>DRAFT</code> & pantau kesehatan platform.	Dashboard 5 KPI → Grafik 14 hari → Moderasi 1 klik → Export CSV → Impersonate untuk debug

CAKUPAN NASIONAL 38 PROVINSI

`lib/provinces.ts` mendefinisikan 38 provinsi (UU 2022): Aceh, Bali, Banten, Bengkulu, DI Yogyakarta, DKI Jakarta, Gorontalo, Jambi, Jawa Barat, Jawa Tengah, Jawa Timur, Kalimantan Barat/Selatan/Tengah/Timur/Utara, Kepulauan Bangka Belitung, Kepulauan Riau, Lampung, Maluku, Maluku Utara, NTB, NTT, Papua, Papua Barat/Barat Daya/Pegunungan/Selatan/Tengah, Riau, Sulawesi Barat/Selatan/Tengah/Tenggara/Utara, Sumatera Barat/Selatan/Utara. Seed menempatkan 23 provider dari Banda Aceh hingga Jayapura — bukan hanya Jawa.

6 Fitur Utama Lengkap (PRD)

6.1 Fitur Publik — Tanpa Login

FITUR	ROUTE / FILE	DETAIL
Landing Page	<code>app/page.tsx</code> force-dynamic	Hero ShieldCheck pill <i>38 provinsi Aceh → Papua</i> + headline <i>orang terpercaya</i> + search besar + pills kategori (2 featured <code>lg:p-6</code> + blur) + 6 jasa terbaru <code>ServiceTile</code> + Cara Kerja 4 langkah (Search/Users/CreditCard/Truck) + Statistik 4 kolom (<code>providerCount</code> , <code>completed</code> , <code>rating</code> , <code>cityCount</code> + Counter animasi) + CTA Punya keahlian? . Data: 8 query <code>Promise.all</code> (categories, services ACTIVE, <code>providerCount</code> , <code>completedOrders</code> , <code>ratingAvg</code> aggregate, <code>distinct city</code> , <code>heroProviders</code>).

Katalog	app/services/ page.client.tsx	Filter: kategori (6), lokasi kota/provinsi datalist 40 (Kepulauan Bangka Belitung), harga Rp chip <100rb 100-300rb 300rb+ , rating ≥4.5/4/3, search multi-kata contains judul/deskripsi/nama provider, sort newest/price/rating, paginasi 12. UX: keepPreviousData + opacity 60% saat refetch (bukan blink), aria-live count, sticky chip top-16 + drawer 85vh, Suspense + CatalogSkeleton .
Detail Jasa	app/service/[slug]/ page.tsx	Server force-dynamic : harga, estimasi hari, provider formatCity Title Case, distribusi bintang, 3 jasa serupa rating desc take 3 .SEO: generateMetadata + JSON-LD Service + AggregateRating , noindex jika ARCHIVED, canonical siteUrl/service/slug .Galeri: aspect-video (mobile 4/3), thumb border-primary shadow-glow , lightbox Dialog max-w-4xl ,Tabs description/reviews/about deep-link ?tab= .
Profil Provider	app/provider/[id]/ page.tsx	Foto pravatar , kota, bio, total jasa aktif, rating agregat, completedOrders , memberSince. List jasa provider.
Search Suggestions	GET /api/search-suggestions	Autocomplete judul jasa + kategori + provider untuk search bar.
SEO & Static	app/sitemap.ts, opengraph-image.tsx, icon.tsx, faq/ page.tsx	Sitemap dinamis, OG image 1200×630, FAQ, robots.

6.2 Fitur Customer — Dari Cari Sampai Ulas

FITUR	ROUTE / FILE	DETAIL
Checkout	app/checkout/ page.tsx	Stepper 2 langkah Detail Jasa → Alamat & Bayar aria-current=step . Field: alamat wajib min 5 char (placeholder nasional <i>Pangkal Pinang, Kepulauan Bangka Belitung</i> + chip), deadline refine masa depan + chips <i>Besok/2 hari/Minggu</i> , metode RadioCardGroup (transfer/ewallet/cod) + PAYMENT_DETAILS , catatan opsional. Ringkasan: subtotal + GRATIS biaya layanan + trust <i>Bisa batal gratis</i> + canvas-confetti saat sukses + wa.me share. Validasi: POST /api/orders cek address required, deadline future, self-order 400 .
Dashboard Customer	app/dashboard/ customer/page.tsx	KPI 4: Total, Berjalan (PENDING+IN_PROGRESS), Selesai, Belanja (! ==CANCELLED). Filter Tabs all/active/completed , list order, SavedServices wishlist.
Order Detail	app/orders/[id]/ page.tsx	OrderTimeline (PENDING → IN_PROGRESS → COMPLETED) + estimasi createdAt+deliveryTimeDays + confetti 80 saat COMPLETED. Panel Informasi: Metode, Tanggal, Alamat, Deadline, Estimasi, Catatan. Aksi: Cancel dialog AlertTriangle (hanya PENDING/IN_PROGRESS), ReviewForm 5 bintang role=radio ArrowLeft/Right roving hanya COMPLETED dalam REVIEW_WINDOW_DAYS 60 , MessageThread max-h-[50vh] polling 8s refetchIntervalInBackground:false + quick chips <i>Haloo kapan bisa?</i>

Wishlist	POST/DELETE /api/saved + SaveButton	Toggle atomik SavedService @@unique([userId,serviceId]) , optimistik + toast. Tampil di dashboard customer.
Ulasan	POST /api/reviews	Rating 1–5 + komentar, recompute computeRatingAggregate transaksional. Update Service.ratingAvg, totalReviews .
Notifikasi	GET /api/notifications + NotificationBell	Bell polling 30s updatedAt , badge count, mark read.

6.3 Fitur Provider — Buka Jasa Sampai Kanban

FITUR	DETAIL
Dashboard Provider app/dashboard/provider/page.tsx	4 StatCard : Total Layanan, Nilai Portofolio (SUM price), Order Masuk (PENDING+IN_PROGRESS), Order Hari Ini (WIB Asia/Jakarta T00:00:00+07:00). Kanban 3 kolom PENDING(warning) → IN_PROGRESS(info) → COMPLETED(success) dot bg-warning/info/success , grip GripVertical drag PointerSensor distance 6 + TouchSensor delay 250 , tombol kontekstual <i>Terima & Kerjakan / Tandai Selesai</i> , guard canTransition + toast <i>Tidak diizinkan: "Menunggu → Selesai"</i> , invalid drop border-destructive/40 , vibrate 10 + confetti , undo hanya jika canTransition(next,prev) . List 6 ServiceTile + Order Terbaru 6 + Revenue & Rating.
Buka Jasa app/dashboard/provider/buka-jasa/page.tsx	Form react-hook-form+zod : title min 3, price >0, deliveryTimeDays 1–90, description 20–500 . Gambar cover dual-mode: Ambil dari Device (drag & drop, accept image/* max 10MB , canvas max 1280 toDataURL jpeg 0.82 auto rasio, preview h-24 + badge <i>Auto-sesuaikan</i>) atau URL https:// (validasi refine terima data:image/), preview kartu langsung bg-accent . Toggle status ACTIVE/DRAFT/ARCHIVED via PATCH /api/services/[id] . Layout flex lg:flex-row self-start shrink-0 (fix kiri ikut naik).
Kelola Pesanan	GET/PATCH /api/provider/orders — update status dengan FSM guard, GET /api/orders/[id]/messages chat.

6.4 Fitur Admin — Moderasi Sampai Export

FITUR	DETAIL
Ringkasan app/dashboard/admin/page.tsx	5 KPI (Counter Rp): Pengguna, Pesanan, Pendapatan (COMPLETED), Jasa Aktif, Rating. Grafik OrdersAreaChart 14 hari + CategoryRevenueChart (dynamic ssr:false). Moderasi DRAFT preview 4 + <i>Buka moderasi</i> , Tabel 8 pesanan terbaru hover:bg-accent , Top 5 Provider formatIDR .
Moderasi app/dashboard/admin/moderasi/page.tsx	PATCH status ACTIVE onMutate busyId + toast + invalidate .
Kelola Pengguna app/dashboard/admin/users/page.tsx	GET /api/admin/users search client, adminRead 60/m , POST /api/admin/users/[id]/impersonate (rate 10/jam) + POST /api/auth/unimpersonate (dual cookie __Secure-), ImpersonationBar .

Export CSV	<code>escapeCsv</code> anti <code>==+\t\r</code> + BOM, <code>adminExport 5/5m</code> , filename <code>orders-YYYY-MM-DD.csv</code> .
<code>GET /api/admin/export</code>	
Stats API	Agregasi <code>groupBy status</code> , aggregate revenue, <code>ordersByDay</code> , <code>categoryRevenue</code> untuk dashboard.
<code>GET /api/admin/stats</code>	

6.5 Fitur Sistem

- **FSM Order** `lib/order-status.ts:17` `PENDING` → `IN_PROGRESS` → `COMPLETED` (terminal) + `CANCELLED` dari `PENDING/IN_PROGRESS` (customer hanya `CANCELLED`). `completedAt` hanya saat `COMPLETED` (jendela ulasan 60 hari tidak reset). Guard di `PATCH /api/orders/[id]` 409 + UI sembunyikan tombol ilegal.
- **Rate Limit Token Bucket** `lib/rate-limit.ts` `MAX 10k` + sweep periodik, per-user `clientKey`, header `RateLimit-*`, 14 rule (register 5/jam, login 10/m, order 20/jam, review 10/jam, impersonate 10/jam, export 5/5m, dll).
- **Upload Device** canvas 1280 jpeg 0.82, fallback URL, `unoptimized` untuk `data:` (Next Image).
- **Seed Nasional** 46 jasa 44 aktif, `galleryPool` per kategori (2 foto tambahan selain foto utama), `addressSamples` 20 kota, deadline future.
- **Dark Mode** penuh — token light/dark terpisah, toggle `ThemeProvider`.

7 User Flow End-to-End

```
[Landing hero 38 provinsi pills "Pangkal Pinang"]
  ↓ klik pill → /services?location=Kepulauan Bangka Belitung
[Catalog filter city contains lower "bangka" → hasil Bangka]
  ↓ klik ServiceTile "Bersih Rumah Bangka"
[Detail /service/[slug]?tab=reviews — galeri 4/3 swipe, Tabs deep-link, formatCity]
  ↓ Pesan Sekarang
[Checkout ?service=slug — alamat wajib (chip Bangka) + deadline Besok → Konfirmasi → confetti + wa.me share]
  ↓ POST /api/orders (address required, deadline future, self-order 400)
[Orders/[id] — OrderTimeline + estimasi createdAt+deliveryTimeDays + chat + cancel dialog]
  ↓ Provider: Dashboard Kanban drag Terima → IN_PROGRESS
[Notification bell 30s (updatedAt) → Provider Tandai Selesai → COMPLETED]
  ↓ Customer: ReviewForm 5★ Arrow keys → POST /api/reviews transaksi
[Service ratingAvg 4.48 recompute, Dashboard Total Belanja update]
```

FSM Guard: `TRANSITIONS = { PENDING: [IN_PROGRESS, CANCELLED], IN_PROGRESS: [COMPLETED, CANCELLED], COMPLETED: [], CANCELLED: [] }`. Customer hanya boleh `CANCELLED`; provider boleh `IN_PROGRESS/COMPLETED`. `Violasi` → 409 + toast.

8 Arsitektur Sistem

```
app/
  (auth)/ login, register
  api/ 20 handler (auth, categories, services, orders, reviews, saved, notifications, admin,
```



```
provider, search-suggestions)
  services/ page.client.tsx (Suspense + keepPreviousData)
  service/[slug]/ page.tsx (force-dynamic, JSON-LD)
  checkout/ page.tsx
  orders/[id]/ page.tsx
  dashboard/ layout.tsx (sidebar 240px) → customer/, provider/, provider/buka-jasa/, admin/, admin/
moderasi/, admin/users/
  provider/[id]/, faq/, sitemap.ts, opengraph-image.tsx

components/
  ui/ button, card, badge, dialog, tabs, input, skeleton, avatar, empty-state, radio-card-group, form
  services/ service-tile, catalog-filters, save-button, rating-stars
  orders/ order-timeline, message-thread, order-status-badge, review-form
  provider/ order-kanban (@dnd-kit), buka-jasa-form, provider-service-list
  dashboard/ stat-card, admin-charts (recharts dynamic), saved-services, CustomerDashboard
  landing/ hero, reveal, counter
  layout/ Navbar, search-bar, notification-bell, mobile-menu, impersonation-bar, page-shell
  three/ hero-orb (R3F 1400 sphere)

lib/
  auth.ts (NextAuth v5 Credentials JWT 30d), prisma.ts, order-status.ts (FSM),
  rate-limit.ts (Token Bucket), api-guard.ts, location.ts (city parser),
  provinces.ts (38), site-url.ts, motion.ts, rating.ts, ids.ts (ObjectId helper), utils.ts

prisma/
  schema.prisma (SQLite dev) + schema.mongo.prisma (MongoDB prod)
  migrations/ 7 + seed.js (46 jasa) + backfill-city.js

tests/ 7 files 108 tests | e2e/ order-lifecycle + public-pages | vercel.json (sin1) | scripts/vercel-
prepare.js
```

9 Model Data — 7 Collections MongoDB

prisma/schema.mongo.prisma (provider mongodb, ObjectId) — production di Atlas ferz/servislokal . Dev tetap SQLite via schema.prisma + switch otomatis scripts/vercel-prepare.js .

MODEL	FIELD PENTING	INDEX
User	id ObjectId, name, email @unique, passwordHash, role CUSTOMER/PROVIDER/ADMIN, location, city, avatarUrl (pravatar), bio, createdAt, updatedAt	@@index([role, city])
Category	id, name @unique, slug @unique, icon, description	—
Service	id, providerId → User, categoryId, title, slug @unique, description, price Float, deliveryTimeDays, imageUrl, images (JSON), ratingAvg, totalReviews, status ACTIVE/DRAFT/ARCHIVED, createdAt	@@index([categoryId,status]), @@index([providerId])
SavedService	userId, serviceId, createdAt	@@unique([userId,serviceId]), @@index([serviceId])

Order	id, customerId, serviceId, totalPrice, status PENDING/IN_PROGRESS/COMPLETED/CANCELLED, paymentMethod, orderNotes, address String (wajib), deadline DateTime?, createdAt, updatedAt, completedAt	@@index([customerId,status]), @@index([serviceId])
Message	orderId, senderId, content, createdAt	@@index([orderId,createdAt])
Review	orderId, reviewerId, rating 1-5, comment, createdAt	—

Kenapa city terpisah? Filter awal pakai `location contains` cocok "Jl" → seluruh katalog. `city` lowercased, indexed, tidak butuh `mode:insensitive` (tidak ada di SQLite, case-sensitive di PG/Mongo). Ditulis saat create/update user & service, bukan dihitung tiap request (scalable).

10 API Endpoints — 20 Handler

METHOD	PATH	AUTH	RATE LIMIT	FUNGSI
POST	/api/auth/register	—	5/jam	Registrasi, hash bcrypt, role default CUSTOMER
POST	/api/auth/ [...nextauth]	—	—	NextAuth v5 Credentials authorize
GET	/api/categories	—	—	List 6 kategori + _count services
GET	/api/services	—	—	List paginated 12, filter q/category/ location/price/rating/sort, hanya ACTIVE
GET	/api/services/[id]	—	—	Detail 1 service + provider + category
PATCH	/api/services/[id]	PROVIDER owner	—	Update service (title/price/desc/status/ images)
DELETE	/api/services/[id]	PROVIDER owner	—	Hapus service (guard: tidak ada order aktif)
GET	/api/services/[id]/ reviews	—	—	List review per service
GET	/api/search- suggestions	—	—	Suggest q → jasa/kategori/provider
GET/POST/ DELETE	/api/saved	CUSTOMER	—	Wishlist toggle
GET/POST	/api/orders	CUSTOMER	20/jam	List my orders / Create order (address wajib, deadline future)

GET/PATCH	/api/orders/[id]	owner/provider	—	Detail / Update status FSM guard 409
GET/POST	/api/orders/[id]/messages	participant	—	Chat polling 8s
GET/PATCH	/api/provider/orders	PROVIDER	—	List & update order milik provider
GET/POST	/api/provider/services	PROVIDER	—	CRUD jasa provider
POST	/api/reviews	CUSTOMER	10/jam	Create review (hanya COMPLETED, window 60h)
GET	/api/notifications	user	—	Polling updatedAt
GET	/api/admin/stats	ADMIN	60/m	Aggregasi KPI + grafik
GET/PATCH	/api/admin/users	ADMIN	60/m	List & update user/role
POST	/api/admin/users/[id]/impersonate	ADMIN	10/jam	Login sebagai user
POST	/api/auth/unimpersonate	ADMIN (impersonating)	—	Kembali ke admin
GET	/api/admin/export	ADMIN	5/5m	Export CSV orders + BOM anti formula

11 Halaman per Halaman — 16 Route

HALAMAN	FILE	HIGHLIGHT UX
/ Landing	app/page.tsx	Force-dynamic 8 query, Reveal + Counter, hero pills 38 provinsi, blur glow disiplin
/services	page.client.tsx	h1 di atas grid (a11y), sidebar sticky top-24 + drawer 85vh + grabber, chip sticky, empty Reset
/service/[slug]	service/[slug]/page.tsx	Metadata + JSON-LD, galeri lightbox, Tabs deep-link, related 3
/provider/[id]	provider/[id]/page.tsx	Stat provider, list jasa
/checkout	checkout/page.tsx	Stepper, RadioCardGroup, PAYMENT_DETAILS, mobile fixed bottom total, confetti
/orders/[id]	orders/[id]/page.tsx	Timeline + estimasi + chat + review, cancel dialog
/dashboard	dashboard/page.tsx	Redirect by role (customer/provider/admin)
/dashboard/customer	dashboard/customer/page.tsx	KPI 4, Tabs all/active/completed, EmptyState
/dashboard/provider		Kanban + StatCard 4 + revenue/rating

	dashboard/provider/ page.tsx	
/dashboard/provider/ buka-jasa	buka-jasa/page.tsx	Form dual-mode image, preview live
/dashboard/admin	admin/page.tsx	5 KPI, 2 chart dynamic, moderasi, Top 5
/dashboard/admin/ moderasi	moderasi/page.tsx	Approve DRAFT → ACTIVE
/dashboard/admin/ users	users/page.tsx	Search + impersonate
/login, /register	(auth)/	Form zod, error CredentialsSignin silent
/faq	faq/page.tsx	Accordion FAQ

Aksesibilitas: 1 h1/halaman, CardTitle as h2, focus-ring, RadioCardGroup roving tabindex Arrow, aria-live, skip-link, scrollbar tipis, kontras AA (47 test color-contrast).

12 Design System — Titik Terang

VISI

Oranye menyala di hitam pekat — hangat tetangga + tegas transaksi. Krem #FBFAF8 bukan putih klinis. Dua mode penuh, bukan dark asal. Referensi: Fiverr (katalog hero), Thumbtack/Urban Company (marketplace lokal), Linear (expo easing, kanban tenang), Vercel/Framer (grid noise + glow disiplin), Stripe (shadow berlapis), Raycast (kbd /), Gojek (KPI uang hero). **Anti-slop:** tanpa gradien ungu, tanpa emoji, tanpa rounded-2xl tanpa shadow.

TOKEN WARNA

TOKEN	LIGHT	DARK	KONTRAS
background	40 33% 98% #FBFAF8	20 14% 4% #0c0a09	—
primary	25 95% 53% #F97316	28 96% 58% #fb923c	—
primary-strong	22 90% 38%	—	5.0:1 AA
border	35 20% 88%	24 10% 16%	—
success	158 74% 27% / soft 94%	—	—
rating	#f59e0b amber	—	—

TIPOGRAFI, RADIUS, SHADOW, MOTION

- Font:** Plus Jakarta Sans + Instrument_Serif via next/font , clamp text-6xl 2.5rem-4.5rem , text-3xl fluid .
- Radius/Shadow:** xl 24px card, shadow 0 8px 24px rgba(12,10,9,.06) light / 0 12px 32px rgba(0,0,0,.5) dark, glow 0 8px 32px rgba(251,122,35,.25) .

- **Motion:** `DURATION fast 150 / base 250 / reveal 800` , `GSAP_EASE smooth power3.out` , `prefers-reduced-motion kill-switch ( globals.css:139 )` , hero orb `useFrame` skip jika reduce, reveal Counter.

13 Keamanan, Performa & Kualitas

- **Rate Limit Token Bucket** `lib/rate-limit.ts` — MAX 10k + sweep periodik, per-user clientKey, header `RateLimit-*` , 14 rule. 401/429 JSON konsisten via `api-guard.ts` .
- **FSM Guard** — `canTransition + transitionError` , 409 jika ilegal, `completedAt` hanya saat COMPLETED.
- **Auth** — NextAuth v5 Credentials JWT 30d, bcrypt 10, CredentialsSignin silent log, dual cookie `__Secure-` untuk impersonate, middleware whitelist + role guard.
- **Headers** — `next.config.mjs` `X-Content-Type-Options nosniff`, `X-Frame-Options DENY`, `HSTS`, `no-store` untuk /api. CSV `escapeCsv` anti formula `=+@\t\r` + BOM.
- **Performa** — `keepPreviousData` katalog, `recharts dynamic ssr:false (~150kB)`, `canvas 1280 jpeg 0.82` untuk upload, `vercel.json sin1` , polling `refetchIntervalInBackground:false` .

14 Teknologi

KATEGORI	STACK	VERSI	ALASAN
Framework	Next.js App Router	15.5.24 (patched CVE-2025-66478)	SSR + RSC + force-dynamic + metadata API
UI	React 19, Tailwind 3.4, Radix, lucide-react	19.0.0 / 3.4.0	Design system + a11y primitive
DB	Prisma 5.20 + MongoDB Atlas (prod) / SQLite (dev)	5.22.0	ObjectId + switch via vercel-prepare.js
Auth	NextAuth v5 Credentials JWT	5.0.0-beta.32	Credentials tanpa OAuth, JWT 30d
State	TanStack Query 5	5.70.0	keepPreviousData + cache
Animasi	GSAP 3.15, R3F 9.7, three 0.185	—	Hero orb 1400 sphere + reveal
Validasi	zod 3.23 + react-hook-form 7.53	—	Form + API guard
Chart	recharts 3.10 (dynamic)	—	Admin KPI
Deploy	Vercel sin1 + MongoDB Atlas ap-southeast-1	—	Latensi rendah Indonesia

15 Pengujian

- `npm run typecheck` 0 error, `npm test` 108 tests: color-contrast 47, order-status 17, rate-limit 13, location 12, rating 6, csv 8, utils 5. `npm run build` 13 route OK.
- `e2e/ order-lifecycle.spec.ts + public-pages.spec.ts` 16 serial via Playwright 1.62.1 .
- CI `.github/workflows/ci.yml` `verify` (migrate+seed+typecheck+unit+build+smoke) + `e2e` (cache browser).

16 Deployment

VERCEL — PRODUCTION LIVE

URL: `https://weblomba-rouge.vercel.app` (primary) + `https://weblomba.vercel.app` . **Project:** `weblomba` (`prj_dIxWAg505regeZJmJ8be64ncm9cI`) , `org` `ferzdevzs-projects` , `region` `sin1` .

```
Build: npm run vercel-build = node scripts/vercel-prepare.js && prisma generate && next build
# vercel-prepare.js: deteksi DATABASE_URL
# - mongodb+srv:// → copy prisma/schema.mongo.prisma → prisma/schema.prisma
# - postgres:// → inject provider postgresql
# - else → keep sqlite
Env Production+Preview (vercel env ls):
  DATABASE_URL (Secret) mongodb+srv://
ferdinandderosaputra200409_db_user:**@ferz.bj80suv.mongodb.net/servislokal?appName=ferz
  AUTH_SECRET / NEXTAUTH_SECRET (Secret) mapagbAyvJ4fEw... (base64 32)
  AUTH_TRUST_HOST=true, AUTH_URL/NEXTAUTH_URL=https://weblomba.vercel.app,
  NEXT_PUBLIC_SITE_URL=https://weblomba.vercel.app
```

MONGODB ATLAS

Cluster **ferz** AWS `ap-southeast-1` `Singapore` , tier Free, version 8.0.29. Database `servislokal` , 7 collections + 5 indexes. Network Access `0.0.0.0/0` (allow Vercel) + `119.235.214.92/32` . User `ferdinandderosaputra200409_db_user` `atlasAdmin` . Seed: `node prisma/seed.js` → 46 jasa (44 ACTIVE), 23 provider, 38 provinsi.

Verifikasi Live: `curl https://weblomba-rouge.vercel.app/api/services` → 200 `{"services":[...]}` 12 per page. `curl https://weblomba-rouge.vercel.app/` → 200 170kB HTML.

17 Dampak, Skalabilitas & Roadmap

DAMPAK TERUKUR

Provider lokal naik estimasi **3–5 order/bulan** (dari sepi) → pendapatan + Rp 500rb–2jt/bulan. Pelanggan banding harga **5 menit vs 1 hari** (katalog vs WA satu-per-satu). Admin pantau kesehatan platform real-time (5 KPI + grafik).

SKALABILITAS

- `city indexed` — filter $O(\log n)$, siap 10k provider.
- Migrasi Prisma 7 + 2 baru — schema versioned.

- `vercel-build sin1` — scale to zero, edge.
- Polling siap ganti **Redis/Pusher/WebSocket** untuk realtime.

ROADMAP 3 SPRINT

SPRINT	FOKUS	DELIVERABLE
Sprint 1 (2 minggu)	Transaksi Real	Midtrans integration, Upstash Redis rate limit distributed, AuditLog collection
Sprint 2 (2 minggu)	Realtime & Discovery	Pusher chat realtime, autocomplete lokasi (Nominatim), peta provider
Sprint 3 (2 minggu)	Ops & Scale	CI Sentry, Neon branch DB, image CDN (Cloudinary), Lighthouse 95+

18 Lampiran

A. AKUN DEMO (PASSWORD: PASSWORD123)

EMAIL	ROLE	KOTA	KEGUNAAN DEMO
<code>admin@servislokal.id</code>	ADMIN	Nasional	Dashboard admin, moderasi, export, impersonate
<code>budi.santoso@email.com</code>	PROVIDER	Jakarta Selatan	Kanban + buka jasa
<code>siti.rahayu@email.com</code>	PROVIDER	Bandung	Kebersihan
<code>agus.pratama@email.com</code>	PROVIDER	Surabaya	Listrik
<code>hendri.bangka@email.com</code>	PROVIDER	Pangkal Pinang	Bangka Belitung — demo nasional
<code>lina.natuna@email.com</code>	PROVIDER	Tanjung Pinang	Kepulauan Riau
<code>jayapura.tangguh@email.com</code>	PROVIDER	Jayapura	Papua
<code>dewi.lestari@email.com</code>	CUSTOMER	Jakarta	Checkout + ulasan

+ 15 provider & 12 customer lain di seed (total 30 user). Lihat `prisma/seed.js:39`.

B. CARA MENJALANKAN LOKAL

```
git clone https://github.com/FerzDevZ/web-lomba.git
cd web-lomba
npm ci
# .env — pilih salah satu:
# SQLite (dev cepat):
DATABASE_URL="file:./dev.db"
# MongoDB Atlas (prod mirror):
```

```
DATABASE_URL="mongodb+srv://
ferdinandderosaputra200409_db_user:avMSsZy8ZF2wFp8R@ferz.bj80suv.mongodb.net/servislokal?
appName=ferz"
AUTH_SECRET="$(openssl rand -base64 32)"
AUTH_TRUST_HOST=true
NEXT_PUBLIC_SITE_URL=http://localhost:3000

npx prisma generate
npx prisma migrate deploy # atau npx prisma db push (MongoDB)
node prisma/seed.js      # 46 jasa + 30 user
npm run dev              # http://localhost:3000
npm run build && npm start # production check
npm test                 # 108 unit
npx playwright test      # e2e
```

C. ENVIRONMENT VARIABLES

KEY	WAJIB	CONTOH	KETERANGAN
DATABASE_URL	Ya	<code>mongodb+srv://.../servislokal</code>	Auto-detect oleh <code>vercel-prepare.js</code>
AUTH_SECRET	Ya	<code>openssl rand -base64 32</code>	NextAuth v5, 32 byte
NEXTAUTH_SECRET	Ya	sama dengan AUTH_SECRET	Alias (kompat)
AUTH_TRUST_HOST	Ya	<code>true</code>	Vercel proxy
AUTH_URL / NEXTAUTH_URL	Ya	<code>https://weblomba.vercel.app</code>	Canonical URL
NEXT_PUBLIC_SITE_URL	Ya	<code>https://weblomba.vercel.app</code>	Untuk sitemap & OG

D. PERINTAH PENTING

```
npm run vercel-build # Vercel: prepare + generate + build
npm run db:seed      # node prisma/seed.js
npm run db:backfill-city # node prisma/backfill-city.js (isi city yang null)
npx prisma generate  # generate client
npx prisma db push --accept-data-loss # MongoDB (tanpa migrate)
npx prisma migrate deploy # SQL (SQLite/Postgres)
./node_modules/.bin/vercel --prod --yes # deploy manual
./node_modules/.bin/vercel env ls      # cek env
```

E. LINK PRODUKSI

- Live: `https://weblomba-rouge.vercel.app` (200 OK, 170kB), `https://weblomba.vercel.app`
- API: `/api/services` (200, 12/page), `/api/categories` (200)
- Repo: `github.com/FerzDevZ/web-lomba` branch `main` (commit `5fe8fd6` fix: upgrade Next.js 15.5.24)
- Atlas: `ferz.bj80suv.mongodb.net` — Network Access 0.0.0.0/0 Active

Dokumen ini di-generate dari codebase aktual `weblomba` pada 30 Agustus 2025 — 30 Agustus 2026.

ServisLokal — Jasa terbaik dari orang terpercaya di sekitarmu. • Dibuat dengan Next.js, Prisma, dan untuk 38 provinsi Indonesia.